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1. Let $(\Omega_1, \mathcal{F}_1, \mu_1)$ be a measure space and let $(\Omega_2, \mathcal{F}_2)$ be a measurable space. Given a measurable function $f : (\Omega_1, \mathcal{F}_1) \to (\Omega_2, \mathcal{F}_2)$, consider the following set function $\mu_2 : \mathcal{F}_2 \to [0, \infty]$ defined by $\mu_2(A) := \mu_1(f^{-1}(A))$, for all $A \in \mathcal{F}_2$.

   (a) Show that μ_2 is a measure.

   (b) Further, for any measurable function $g : (\Omega_2, \mathcal{F}_2) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$, we have for all $A \in \mathcal{F}_2$

$$\int_A g d\mu_2 = \int_{f^{-1}(A)} (g \circ f) d\mu_1$$

provided any one of the integrals exist.

Step 1:- Indicator function

$$\boxed{\mu_2(A \cap B) = \mu_1(f^{-1}(A \cap B)) = \mu_1(f^{-1}(A) \cap f^{-1}(B))}$$

Let $g = I_B$, $B \in \mathcal{F}_2$.

Then $\int_A g \, d\mu_2 = \int_\Omega g I_A \, d\mu_2 = \int_\Omega I_{A \cap B} \, d\mu_2 = \mu_2(A \cap B)$

$= \mu_1(f^{-1}(A \cap B))$

$= \mu_1(f^{-1}(A) \cap f^{-1}(B))$

$\int_{f^{-1}(A)} (g \circ f) \, d\mu_1 = \int_{f^{-1}(A)} I_{f^{-1}(B)} \, d\mu_1 = \int_\Omega I_{f^{-1}(A) \cap f^{-1}(B)} \, d\mu_1$

$= \mu_1(f^{-1}(A) \cap f^{-1}(B))$

Step 2:- Simple function

$g = \sum_{i=1}^{n} x_i I_{B_i}$

LHS $= \int_A g \, d\mu_2 = \int_\Omega g I_A \, d\mu_2$

$$\boxed{g I_A = \sum_{i=1}^{n} x_i I_{A \cap B_i}}$$

$\therefore \int_\Omega g I_A \, d\mu_2 = \sum_{i=1}^{n} x_i \mu_2(A \cap B_i) = \sum_{i=1}^{n} x_i \mu_1(f^{-1}(A \cap B_i))$.

RHS $g \circ f = \sum_{i=1}^{n} x_i I_{B_i} \circ f = \sum_{i=1}^{n} x_i I_{f^{-1}(B_i)}$

$\int_{f^{-1}(A)} g \circ f \, d\mu_1 = \int_\Omega (g \circ f) I_{f^{-1}(A)} \, d\mu_1 = \sum_{i=1}^{n} x_i \mu_1(f^{-1}(A) \cap f^{-1}(B_i))$

$$\int_{f^{-1}(A)} g\circ f \, d\mu_1 = \int_{\Omega} (g\circ f)\, I_{f^{-1}(A)} \, d\mu_1 = \sum_{i=1}^{n} x_i \, \mu_1\left(f^{-1}(A) \cap f^{-1}(B_i)\right)$$

$$= \sum_{i=1}^{n} x_i \, \mu_1\left(f^{-1}(A \cap B_i)\right)$$

Step-3: (Non-negative measurable function)

Let g be any non negative measurable function. Then there exists sequence of non negative simple functions s_n such that $s_n \uparrow g$.

$\therefore$ By MCT $\lim_{n\to\infty} \int_A s_n \, d\mu_2 = \int_A g \, d\mu_2$

$$\int_A g \, d\mu_2 = \lim_{n\to\infty} \int_A s_n \, d\mu_2 = \lim_{n\to\infty} \int_{f^{-1}(A)} s_n \circ f \, d\mu_1 = \int_{f^{-1}(A)} g\circ f \, d\mu_1$$

(By MCT)

Step-4: Arbitrary measurable function

Let g be any arbitrary measurable function

$$\int_A g \, d\mu_2 = \int_A g^+ d\mu_2 - \int_A g^- d\mu_2 = \int_{f^{-1}(A)} g^+ \circ f \, d\mu_1 - \int_{f^{-1}(A)} g^- \circ f \, d\mu_1$$

$$= \int_{f^{-1}(A)} g\circ f \, d\mu_1$$

2. Let $f : (\mathbb{R}, \mathcal{B}(\mathbb{R})) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ be measurable and fix $a \in \mathbb{R}$. Then

$$\int_{\mathbb{R}} f \, d\delta_a = f(a).$$

→ Step-I :- $f = I_B$ $\quad \int_{\mathbb{R}} I_B \, d\delta_a = \delta_a(B) = \begin{cases} 1, & a \in B \\ 0, & a \notin B \end{cases}$

$$= I_B(a)$$

$$= f(a)$$

Step-II :- $f = \sum_{i=1}^{n} c_i I_{B_i}$

<u>step - II</u> :- $f = \sum_{i=1}^{n} c_i I_{B_i}$

$$\int_{\mathbb{R}} f \, \delta_a = \sum_{i=1}^{n} c_i \, \delta_a(B_i) = \sum_{i=1}^{n} c_i I_{B_i}(a) = f(a)$$

<u>step - III</u>: f is non-negative measurable.

$f_n \uparrow f$

$\downarrow$

non-negative simple function

$$\int_{\mathbb{R}} f \, \delta_a = \lim_{n\to\infty} \int f_n \, \delta_a = \lim_{n\to\infty} f_n(a) = f(a)$$

<u>step - IV</u>: f is arbitrary measurable function

$$\int_{\mathbb{R}} f \, \delta_a = \int_{\mathbb{R}} f^+ \delta_a - \int_{\mathbb{R}} f^- \delta_a = f^+(a) - f^-(a) = f(a)$$

3. Let $\mu_1, \mu_2, \ldots$ be measures on a measurable space $(\Omega, \mathcal{F})$. Then for any measurable function $f : (\Omega, \mathcal{F}) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ we have

- $\int_\Omega f d(\alpha\mu) = \alpha \int_\Omega f d\mu$ for all $\alpha > 0$ (provided any one of the integrals exist), and
- $\int_\Omega f d(\sum_{n=1}^{\infty} \mu_n) = \sum_{n=1}^{\infty} \int_\Omega f d\mu_n$ (provided the integrals on the RHS exists and the sum is well-defined).

→ <u>step - I</u>: f is simple function,

$f = \sum_{i=1}^{n} c_i I_{B_i}$

$$\int_\Omega f \, d(\alpha\mu) = \sum_{i=1}^{n} c_i (\alpha\mu)(B_i) = \sum_{i=1}^{n} \alpha c_i \mu(B_i) = \alpha \sum_{i=1}^{n} c_i \mu(B_i)$$

$$= \alpha \int_\Omega f \, d\mu$$

- <u>step - I</u> :- f is simple function

$f = \sum_{i=1}^{n} c_i I_{B_i}$.
$$\int_\Omega f \, d\left(\sum_{n=1}^{\infty} \mu_n\right) = \sum_{i=1}^{n} c_i \left(\sum_{i=1}^{\infty} \mu_n (B_i)\right)$$

$$=$$

4. Let $X : (\Omega, \mathcal{F}, \mathbb{P}) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ be a discrete RV and let $f : (\mathbb{R}, \mathcal{B}(\mathbb{R})) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ be measurable.

(a) Show that $Y := f(X)$ is a discrete RV.

(b) Write down the law $\mathbb{P} \circ Y^{-1}$ of Y in terms of f and $\mathbb{P} \circ X^{-1}$.

(c) Show that $\int_{\mathbb{R}} y d\mathbb{P} \circ Y^{-1} = \int_{\mathbb{R}} f d\mathbb{P} \circ X^{-1}$ provided one of the integrals exists.

(a) Given X is a discrete rv.

$\therefore \exists$ a countable Borel set S such that $\mathbb{P} \circ X^{-1}(S) = 1$.

$S' = f(S) = \{f(x) : x \in S\}$

Since S is countable and f is a function, S' is also countable set. As f maps to $\mathbb{R}$ and countable subsets of $\mathbb{R}$ are Borel sets, S' is a countable Borel set.

$$\mathbb{P} \circ Y^{-1}(S') = \mathbb{P}(Y \in S') = \mathbb{P}(f(X) \in T) = \mathbb{P}(X \in f^{-1}(T))$$
$$= \mathbb{P}(X \in S)$$
$$= 1$$

$\therefore Y$ is a discrete RV.

(b)
$$\mathbb{P} \circ Y^{-1}(B) = \mathbb{P}(Y \in B) = \mathbb{P}(f(X) \in B)$$
$$= \mathbb{P}(X \in f^{-1}(B))$$
$$= \mathbb{P} \circ X^{-1}(f^{-1}(B))$$
$$= \sum_{x_j \in f^{-1}(B)} \mathbb{P} \circ X^{-1}(\{x_j\}) \, \delta_{x_j}$$

(c) $\int_{\mathbb{R}} y \, d\mathbb{P} \circ Y^{-1} = \int_{\mathbb{R}} f \, d\mathbb{P} \circ X^{-1}$ (s.T) [provided one integral exists]

$$\int_{\mathbb{R}} y \, d\mathbb{P} \circ Y^{-1} = \sum_{y_j \in S'} y_j \, \mathbb{P} \circ Y^{-1}(\{y_j\})$$
$$= \sum_{y_j \in S'} \left(y_j \sum_{x_k \in f^{-1}(\{y_j\})} \mathbb{P} \circ X^{-1}(\{x_k\}) \, \delta_{x_k} \right)$$